Fluid model forgetting case

Standalone theoretical note

This note presents a proposition where we can characterize the region where there is at least one threshold if the set is non empty.

Recall from the previous note

Model and notation. Fix $\rho \in (0,1)$, $\gamma \in (0,1)$, $p_0 \in (0,1)$, $0 < p_1 < p_2 < 1$, and $0 \le c_1 < c_2 < R$. Write $\Delta c = c_2 - c_1$ and $\Delta p = p_2 - p_1$. As in the manuscript, the normalized state space and outcome transitions are

$$\begin{aligned} \mathcal{Z} &= \{(x_s, x_f) : x_s \ge 0,\ x_f \ge 0,\ x_s + x_f \le 1\}, \\ T_\rho^s(x_s, x_f) &= (\rho x_s + 1 - \rho, \rho x_f), \\ T_\rho^f(x_s, x_f) &= (\rho x_s, \rho x_f + 1 - \rho). \end{aligned}$$

Proposition 1 (Characterization of positive advantage)**.** *Let U_ρ^1 be the value of always choosing product 1. For $\rho \in (0,1)$, we have $\mathcal{A}_\rho > 0$ if and only if*

$$\max_{z \in \mathcal{K}_\rho} \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right] > \frac{\Delta c}{\gamma \Delta p}. \tag{1}$$

Proof. Conditional on a purchase from seller A, let

$$Q_\rho^i(z) = R - c_i + \gamma \left[p_i U_\rho^1(T_\rho^s z) + (1 - p_i) U_\rho^1(T_\rho^f z) \right], \qquad i \in \{1, 2\}.$$

This is the value of choosing product i for the current purchase and product 1 thereafter. Subtracting the two expressions gives

$$Q_\rho^2(z) - Q_\rho^1(z) = -\Delta c + \gamma \Delta p \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right].$$

Define

$$\mathcal{A}_\rho^1 = \max_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right] \right\}.$$

The fixed-policy operator is a contraction preserving continuity, so U_ρ^1 is continuous and this maximum exists. We first show that $\mathcal{A}_\rho > 0$ if and only if $\mathcal{A}_\rho^1 > 0$, by proving that their nonpositivity conditions are equivalent.

Suppose $\mathcal{A}_\rho^1 \le 0$. Then $Q_\rho^1(z) \ge Q_\rho^2(z)$ at every $z \in \mathcal{K}_\rho$. By the fixed-policy Bellman equation,

$$U_\rho^1(z) = \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} Q_\rho^1(z) = \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \max_{i \in \{1,2\}} Q_\rho^i(z).$$

Thus U_ρ^1 also satisfies the optimal Bellman equation on $\mathcal{K}_\rho$. Since this set is invariant under both transitions, the Bellman operator restricted to it is a contraction. Uniqueness therefore gives $U_\rho^1 = V_\rho$ on $\mathcal{K}_\rho$, and hence $\mathcal{A}_\rho = \mathcal{A}_\rho^1 \le 0$.

Conversely, suppose $\mathcal{A}_\rho \leq 0$. Product 1 then maximizes the Bellman expression evaluated at V_ρ at every state in $\mathcal{K}_\rho$. Consequently, V_ρ satisfies the Bellman equation for always choosing product 1 on this invariant set. Uniqueness of the fixed-policy value gives $V_\rho = U_\rho^1$ there, and hence $\mathcal{A}_\rho^1 = \mathcal{A}_\rho \leq 0$.

We have proved $\mathcal{A}_\rho \leq 0$ if and only if $\mathcal{A}_\rho^1 \leq 0$. Taking negations gives the desired equivalence of strict positivity. Finally, since Δc and $\gamma \Delta p > 0$ do not depend on z,

$$\mathcal{A}_\rho^1 = -\Delta c + \gamma \Delta p \max_{z \in \mathcal{K}_\rho} \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right].$$

Therefore $\mathcal{A}_\rho^1 > 0$ is equivalent to the inequality in the statement. □

Now we can think about the information that the fix policy of using product 2 gives us. Only because it is easier to read let's have the following notation

$$a_\rho(z) = -\Delta c + \gamma \Delta p \left[V_\rho(T_\rho^s z) - V_\rho(T_\rho^f z) \right] \tag{2}$$

$$a_\rho^1(z) = -\Delta c + \gamma \Delta p \left[U_\rho^1(T_\rho^s z) - U_\rho^1(T_\rho^f z) \right] \tag{3}$$

$$a_\rho^2(z) = -\Delta c + \gamma \Delta p \left[U_\rho^2(T_\rho^s z) - U_\rho^2(T_\rho^f z) \right] \tag{4}$$

Then $\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho} a_\rho(z)$ and $\mathcal{A}_\rho^1 = \max_{z \in \mathcal{K}_\rho} a_\rho^1(z)$. We define

$$\mathcal{A}_\rho^2 = \min_{z \in \mathcal{K}_\rho} a_\rho^2(z) \tag{5}$$

We can characterize the region where there exist at least one threshold thinking about the intersection of states where there is used of product 2 based on conditions determined over $\mathcal{A}_\rho^1$ and $\mathcal{A}_\rho^2$.

Proposition 2. *For $\rho \in (\frac{1}{2}, 1)$, if $\mathcal{A}_\rho^1 > 0$ and $\mathcal{A}_\rho^2 < 0$, then there exists $z^* \in \mathcal{K}_\rho$ such that $a_\rho(z^*) = 0$.*

Proof. By proposition 1 we have that $\mathcal{A}_\rho^1 > 0$ implies that there exist $z_1 \in \mathcal{K}_\rho$ such that $a_\rho(z_1) > 0$. We can prove that $\mathcal{A}_\rho^2 < 0 \iff \min_{z \in \mathcal{K}_\rho} a_\rho(z) < 0$. Indeed, suppose that $\mathcal{A}_\rho^2 \geq 0$ then, for all $z \in \mathcal{K}_\rho$

$$-\Delta c + \gamma \Delta p \left[U_\rho^2(T_\rho^s z) - U_\rho^2(T_\rho^f z) \right] \geq 0 \tag{6}$$

Hence, product 2 maximizes the Bellman expression evaluated at U_ρ^2 at every state in $\mathcal{K}_\rho$. By the fixed-policy Bellman equation,

$$\begin{aligned} U_\rho^2(z) &= \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \left\{ R - c_2 + \gamma \left[p_2 U_\rho^2(T_\rho^s z) + (1 - p_2) U_\rho^2(T_\rho^f z) \right] \right\} \\ &= \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \max_{i \in \{1,2\}} \left\{ R - c_i + \gamma \left[p_i U_\rho^2(T_\rho^s z) + (1 - p_i) U_\rho^2(T_\rho^f z) \right] \right\}. \end{aligned}$$

Thus U_ρ^2 also satisfies the optimal Bellman equation on $\mathcal{K}_\rho$. Since this set is invariant under both transitions, the Bellman operator restricted to it is a contraction. Uniqueness therefore gives $U_\rho^2 = V_\rho$ on $\mathcal{K}_\rho$. Both successor states remain in this set, so $a_\rho^2 = a_\rho$ there, and hence

$$\min_{z \in \mathcal{K}_\rho} a_\rho(z) = \min_{z \in \mathcal{K}_\rho} a_\rho^2(z) = \mathcal{A}_\rho^2 \geq 0.$$

Conversely, suppose $\min_{z\in\mathcal{K}_\rho} a_\rho(z) \geq 0$. Then $a_\rho(z) \geq 0$ at every state in $\mathcal{K}_\rho$, so product 2 maximizes the Bellman expression evaluated at V_ρ. Consequently,

$$V_\rho(z) = \frac{D_\rho(z)}{1-\gamma[1-D_\rho(z)]}\left\{R - c_2 + \gamma\left[p_2 V_\rho(T_\rho^s z) + (1-p_2)V_\rho(T_\rho^f z)\right]\right\}.$$

Thus V_ρ satisfies the Bellman equation for always choosing product 2 on this invariant set. Uniqueness of the fixed-policy value gives $V_\rho = U_\rho^2$ there. Therefore $a_\rho = a_\rho^2$ on $\mathcal{K}_\rho$, and

$$\mathcal{A}_\rho^2 = \min_{z\in\mathcal{K}_\rho} a_\rho^2(z) = \min_{z\in\mathcal{K}_\rho} a_\rho(z) \geq 0.$$

We have proved that $\mathcal{A}_\rho^2 \geq 0$ if and only if $\min_{z\in\mathcal{K}_\rho} a_\rho(z) \geq 0$. Taking negations gives

$$\mathcal{A}_\rho^2 < 0 \quad\Longleftrightarrow\quad \min_{z\in\mathcal{K}_\rho} a_\rho(z) < 0.$$

Since $\mathcal{A}_\rho^2 < 0$, there exists $z_2 \in \mathcal{K}_\rho$ such that $a_\rho(z_2) < 0$. Together with $a_\rho(z_1) > 0$, this gives two states at which a_ρ has opposite signs.

Since $\rho > 1/2$, the set $\mathcal{K}_\rho$ is the segment $\{(x_s, 1-x_s) : x_s \in [0,1]\}$. We can therefore write $z_1 = (x_1, 1-x_1)$ and $z_2 = (x_2, 1-x_2)$, with $x_1, x_2 \in [0,1]$. Then

$$a_\rho(x_2, 1-x_2) < 0 < a_\rho(x_1, 1-x_1).$$

Since $x \mapsto a_\rho(x, 1-x)$ is continuous on $[0,1]$, the intermediate value theorem therefore gives

$$x^* \in \big(\min\{x_1, x_2\}, \max\{x_1, x_2\}\big) \subseteq (0,1)$$

such that

$$a_\rho(x^*, 1-x^*) = 0.$$

Thus $z^* = (x^*, 1-x^*) \in \mathcal{K}_\rho$ satisfies $a_\rho(z^*) = 0$. At this state, both products give the same Bellman value and are therefore optimal. □

A cost interval for an interior indifference state. For $i \in \{1,2\}$, let

$$H_\rho^i(z) = \mathbb{E}_z^i\left[\sum_{t=1}^{\infty} \gamma^{t-1}\mathbf{1}\{y_t = A\}\right] \tag{7}$$

be the expected discounted number of purchases from seller A when product i is always used. Here t counts calendar periods, and $\mathbb{E}_z^i$ denotes expectation starting from z under this fixed policy.

Proposition 3 (Cost region with both products strictly optimal at some states)**.** *Fix $\rho \in (1/2, 1)$, $\gamma \in (0,1)$, $p_0 \in (0,1)$, $0 < p_1 < p_2 < 1$, and $0 \leq c_1 < R$. Define*

$$\underline{c}_2 = \frac{c_1 + \gamma\Delta p\, R \min_{z\in\mathcal{K}_\rho}\left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z)\right]}{1 + \gamma\Delta p \min_{z\in\mathcal{K}_\rho}\left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z)\right]},$$

$$\overline{c}_2 = c_1 + \gamma\Delta p\,(R - c_1)\max_{z\in\mathcal{K}_\rho}\left[H_\rho^1(T_\rho^s z) - H_\rho^1(T_\rho^f z)\right].$$

These bounds do not depend on c_2. For any $c_2 \in (c_1, R)$,

$$\mathcal{A}_\rho^1 > 0 \text{ and } \mathcal{A}_\rho^2 < 0 \quad\Longleftrightarrow\quad \underline{c}_2 < c_2 < \overline{c}_2. \tag{8}$$

Thus $(\underline{c}_2, \overline{c}_2) \cap (c_1, R)$ is exactly the cost region where each product is strictly optimal at some state in $\mathcal{K}_\rho$. For every cost in this region, there exists $x^* \in (0,1)$ such that $a_\rho(x^*, 1-x^*) = 0$.

Proof. Under the fixed policy of always choosing product i, the distribution of purchases and outcomes does not depend on costs. Moreover,

$$0 \le H_\rho^i(z) \le \sum_{t=1}^{\infty} \gamma^{t-1} = \frac{1}{1-\gamma} < \infty.$$

Since each purchase gives the constant profit $R - c_i$, we have

$$U_\rho^i(z) = (R - c_i) H_\rho^i(z), \qquad i \in \{1, 2\}.$$

In particular, both H_ρ^1 and H_ρ^2 are independent of c_2.

We first check that the denominator defining $\underline{c}_2$ is positive. Conditioning on the first calendar period gives

$$H_\rho^i(z) = \frac{D_\rho(z)}{1 - \gamma[1 - D_\rho(z)]} \left\{ 1 + \gamma \left[p_i H_\rho^i(T_\rho^s z) + (1 - p_i) H_\rho^i(T_\rho^f z) \right] \right\}.$$

On the boundary, demand $D_\rho(x, 1-x)$ and both successor coordinates are increasing in x. Iteration of this fixed-policy equation from zero therefore shows that $H_\rho^i(x, 1-x)$ is nondecreasing. It follows that

$$H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \ge 0, \qquad z \in \mathcal{K}_\rho,$$

so the denominator defining $\underline{c}_2$ is at least one. The same fixed-policy operator is a contraction preserving continuity, so the extrema in the definitions exist.

Using $\Delta c = c_2 - c_1$ and the factorization of U_ρ^1, we obtain

$$\begin{aligned} \mathcal{A}_\rho^1 &= c_1 - c_2 + \gamma \Delta p \, (R - c_1) \max_{z \in \mathcal{K}_\rho} \left[H_\rho^1(T_\rho^s z) - H_\rho^1(T_\rho^f z) \right] \\ &= \overline{c}_2 - c_2. \end{aligned}$$

Hence $\mathcal{A}_\rho^1 > 0$ if and only if $c_2 < \overline{c}_2$.

Similarly, the factorization of U_ρ^2 gives

$$\mathcal{A}_\rho^2 = c_1 - c_2 + \gamma \Delta p \, (R - c_2) \min_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right].$$

Collecting the terms containing c_2, the condition $\mathcal{A}_\rho^2 < 0$ is equivalent to

$$\begin{aligned} c_1 + \gamma \Delta p \, R \min_{z \in \mathcal{K}_\rho} &\left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right] \\ &< c_2 \left\{ 1 + \gamma \Delta p \min_{z \in \mathcal{K}_\rho} \left[H_\rho^2(T_\rho^s z) - H_\rho^2(T_\rho^f z) \right] \right\}. \end{aligned}$$

Dividing by the positive expression in braces gives $\mathcal{A}_\rho^2 < 0$ if and only if $c_2 > \underline{c}_2$. Combining the two inequalities proves the stated equivalence.

By Proposition 1 and the equivalence proved in Proposition 2, these conditions hold exactly when a_ρ is positive at some state and negative at another. Proposition 2 then gives the interior indifference state. □

Remark. The remaining parameters are arbitrary within the assumptions, but they determine the two cost bounds. The admissible cost region is their open interval intersected with (c_1, R), whenever this intersection is nonempty. The proposition establishes the existence of an interior indifference state; it does not determine the number or ordering of policy switches.